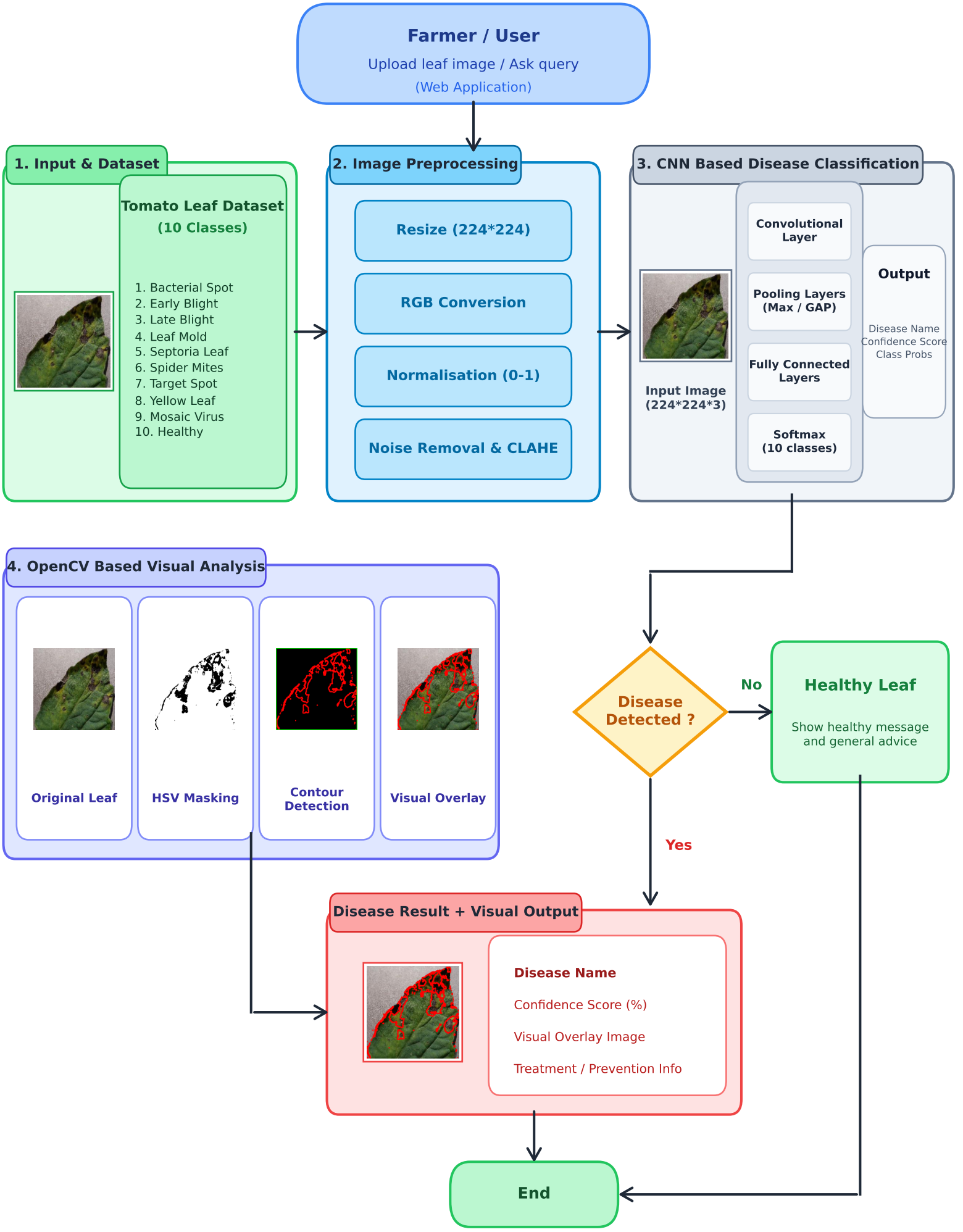

Farmer / User
Upload leaf image / Ask query
(Web Application)
1. Input & Dataset
Tomato Leaf Dataset
(10 Classes)
1. Bacterial Spot
2. Early Blight
3. Late Blight
4. Leaf Mold
5. Septoria Leaf
6. Spider Mites
7. Target Spot
8. Yellow Leaf
9. Mosaic Virus
10. Healthy
2. Image Preprocessing
Resize (224*224)
RGB Conversion
Normalisation (0-1)
Noise Removal & CLAHE
3. CNN Based Disease Classification
Input Image
(224*224*3)
Convolutional Layer
Pooling Layers (Max / GAP)
Fully Connected Layers
Softmax (10 classes)
Output
Disease Name
Confidence Score
Class Probs
4. OpenCV Based Visual Analysis
Original Leaf
HSV Masking
Contour Detection
Visual Overlay
Disease Detected ?
No
Healthy Leaf
Show healthy message and general advice
Yes
Disease Result + Visual Output
Disease Name
Confidence Score (%)
Visual Overlay Image
Treatment / Prevention Info
End